



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2023/2024

**SECI1143 – KEBARANGKALIAN STATISTIK & ANALISIS DATA
(PROBABILITY & STATISTICAL DATA ANALYSIS)**

SECTION 03

PROJECT 2

LECTURER: DR. NOR HAIZAN BT MOHAMED RADZI

STUDENT NAME	MATRIC NO
LAU YEE WEN	A23CS0099
GUI KAH SIN	A23CS0080
CHERYL CHEONG KAH VOON	A23CS0060
TAN SZE QING	A23CS5048

Table of Contents

1.0 Introduction.....	3
2.0 DataSet.....	4
3.0 Data Analysis.....	7
3.1 One-Sample Hypothesis Test.....	7
3.2 Correlation Test.....	8
3.3 Regression Test.....	12
3.4 Chi-Square test of independence.....	17
4.0 Conclusion.....	19
5.0 Appendix.....	20

1.0 Introduction

Heart attacks, or myocardial infarctions, continue to be one of the most significant global health issues, representing a substantial portion of morbidity and mortality rates worldwide. Understanding the factors that contribute to heart attack risks is crucial for developing effective prevention and management strategies. In this project, we aim to investigate the various predictors of heart attack risks by utilising statistical analysis methods on a comprehensive dataset.

The dataset we will use is the Heart Attack Risk Prediction Dataset, sourced from Kaggle. This dataset, collected by SOURAV BANERJEE, a Senior Data Scientist at Launchpad, Kolkata, West Bengal, India, includes a wealth of information from over 8700 patients around the globe (only 1000 actual samples taken). It provides detailed attributes such as age, gender, cholesterol levels, blood pressure, smoking habits, exercise patterns, dietary preferences, and more. By analysing this data, we seek to uncover the complex interplay of these variables and their contribution to heart attack risks.

This study will employ various statistical analysis techniques, including two-sample hypothesis tests, correlation and regression tests, and chi-square test of independence. Our goal is to determine whether the selected variables are dependent on each other and to identify significant predictors of heart attack risks.

The primary objectives of this project are:

- To apply and perform statistical test analysis on the secondary data source.
- To determine the dependency relationships among the selected variables in the dataset.

By achieving these objectives, we hope to contribute valuable insights into the predictors of heart attack risks, ultimately aiding in the development of proactive strategies for cardiovascular disease prevention and management.

2.0 DataSet

Description of data

Population: Patients around the global

Sample: 1000 Patients

Variable	Type of variable	Measurement level
Patient id	Qualitative	Nominal
Age (age of patient)	Quantitative	Ratio
Sex (gender of patient)	Qualitative	Nominal
Cholesterol (cholesterol level of the patient)	Quantitative	Ratio
Blood pressure (blood pressure of the patient)	Quantitative	Ratio
Heart rate (heart rate of the patient)	Quantitative	Ratio
Diabetes (whether the patient has diabetes)	Qualitative	Nominal
Family History (family history of heart-related problems)	Qualitative	Nominal
Smoking (smoking status of the patient)	Qualitative	Nominal
Obesity (obesity status of the patient)	Qualitative	Nominal
Alcohol Consumption (level of alcohol consumption by the patient)	Qualitative	Ordinal
Exercise Hours Per Week (number of exercise hours per week)	Quantitative	Ratio
Diet (dietary habits of the patient)	Qualitative	Ordinal

Previous Heart Problems (previous heart problems of the patient)	Qualitative	Nominal
Medication Use (medication usage by the patient)	Qualitative	Nominal
Stress Level (stress level reported by the patient)	Quantitative	Ordinal
Sedentary Hours Per Day (hours of sedentary activity per day)	Quantitative	Ratio
Income (income level of the patient)	Quantitative	Ratio
BMI (Body Mass Index (BMI) of the patient)	Quantitative	Ratio
Triglycerides (triglyceride levels of the patient)	Quantitative	Ratio
Physical Activity Days Per Week (days of physical activity per week)	Quantitative	Ratio
Sleep Hours Per Day (hours of sleep per day)	Quantitative	Ratio
Country (country of the patient)	Qualitative	Nominal
Continent (continent where the patient resides)	Qualitative	Nominal
Hemisphere (hemisphere where the patient resides)	Qualitative	Nominal
Heart Attack Risk (presence of heart attack risk)	Qualitative	Nominal

Statistical Test Analysis

Selected Variables	Objectives	Test Analysis and Expected Outcome
Age	To determine whether the mean age exceeds 50 or not	Analysis: 1 sample hypothesis test (Test on mean, variance unknown) Possible outcome: By conducting a one-sample t-test, we can determine whether the mean of the patient's age is greater than 50. If the test statistic is greater than the test critical at 0.05 significance level, it indicates that the mean of the patient's age is greater than 50 years old.
Age, Cholesterol	To examine the relationship between age and total cholesterol levels.	Analysis: Correlation analysis Possible Outcome: By calculating the correlation coefficient between age and total cholesterol levels, we can determine the strength and direction of the relationship. If the correlation coefficient is positive and statistically significant, it would indicate that as age increases, total cholesterol levels tend to be higher.
BMI, Heart rate	To investigate whether heart rate is a predictor of BMI	Analysis: Simple linear regression Possible outcome:

		By carrying out a simple linear regression analysis, we are able to determine if there is a relationship between heart rate and BMI. If the regression coefficient is statistically significant with $p < 0.05$, it provides insight that we can quantify the extent of this relationship through this equation of regression.
Sex, Obesity	To determine whether sex affects the occurrence of obesity	Analysis: Chi-Square test of independence Possible outcome: By conducting a chi-square test of independence, we can determine if there is a relationship between the sex and obesity. If the chi-square test falls within an accepted region at a significance level of $\alpha = 0.05$, it could be concluded that the sex and risk of obesity are independent.

3.0 Data Analysis

3.1 One-Sample Hypothesis Test

In this analysis, we will use variable age, where we will test whether the mean of the patient's age exceeds 50 or not at 95% confidence level. The sample size of the database is 1000. The hypothesis test on mean for 1 sample and the population variance is unknown.

Hypothesis Statement:

$$H_0 : \mu=50$$

$$H_1 : \mu>50 \text{ (claim)}$$

Where the μ is represents the population mean of the patient's age.

Test Statistic:

Given a 95% confidence level, $\alpha=0.05$. The test statistic, t_0 can be calculated using the formula:

$$T_0 = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

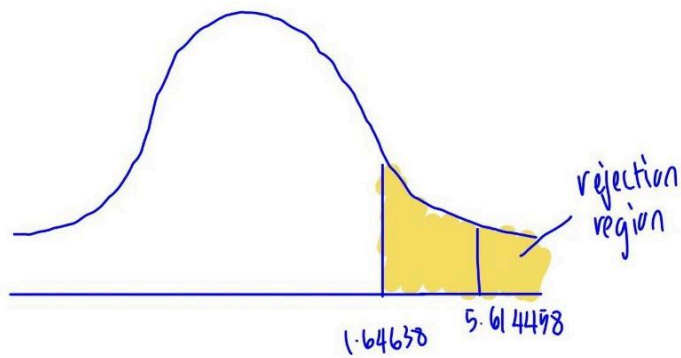
By using RStudio, the mean, $\bar{x}$ and standard deviation, s of the patient's age have been calculated. $\bar{x} = 53.857$, $s = 21.7241$

xbar	53.857
s	21.7240992330519

Using RStudio, the test statistic, t_0 , test critical, $t_{0.05,1000}$ and the probability value, pval have been calculated. $t_0 = 5.614458$, $t_{0.05,1000} = 1.64638$ and $pval = 1.27721e-08$

t	5.61445830477177
t.alpha	1.64638034542753
pval	1.27721029688954e-08

Conclusion:



Since $t_0 = 5.614458 > t_{0.05, 1000} = 1.64638$, we **reject** $H_0: \mu=50$ at the 0.05 significance level. There is enough evidence to support the claim that the mean of the patient's age is greater than 50 years old.

3.2 Correlation Test

In this analysis, we will use the variables **age** and **Cholesterol** to determine whether there is a linear relationship between age and cholesterol level of the patient at a 95% confidence level in a sample size of 1000 using Pearson's Product-Moment Correlation Coefficient. The strength of association, which is the linear relationship between two variables, is measured using correlation analysis. Since both of our data are ratio type, Pearson's technique was selected for calculating the sample correlation coefficient, r .

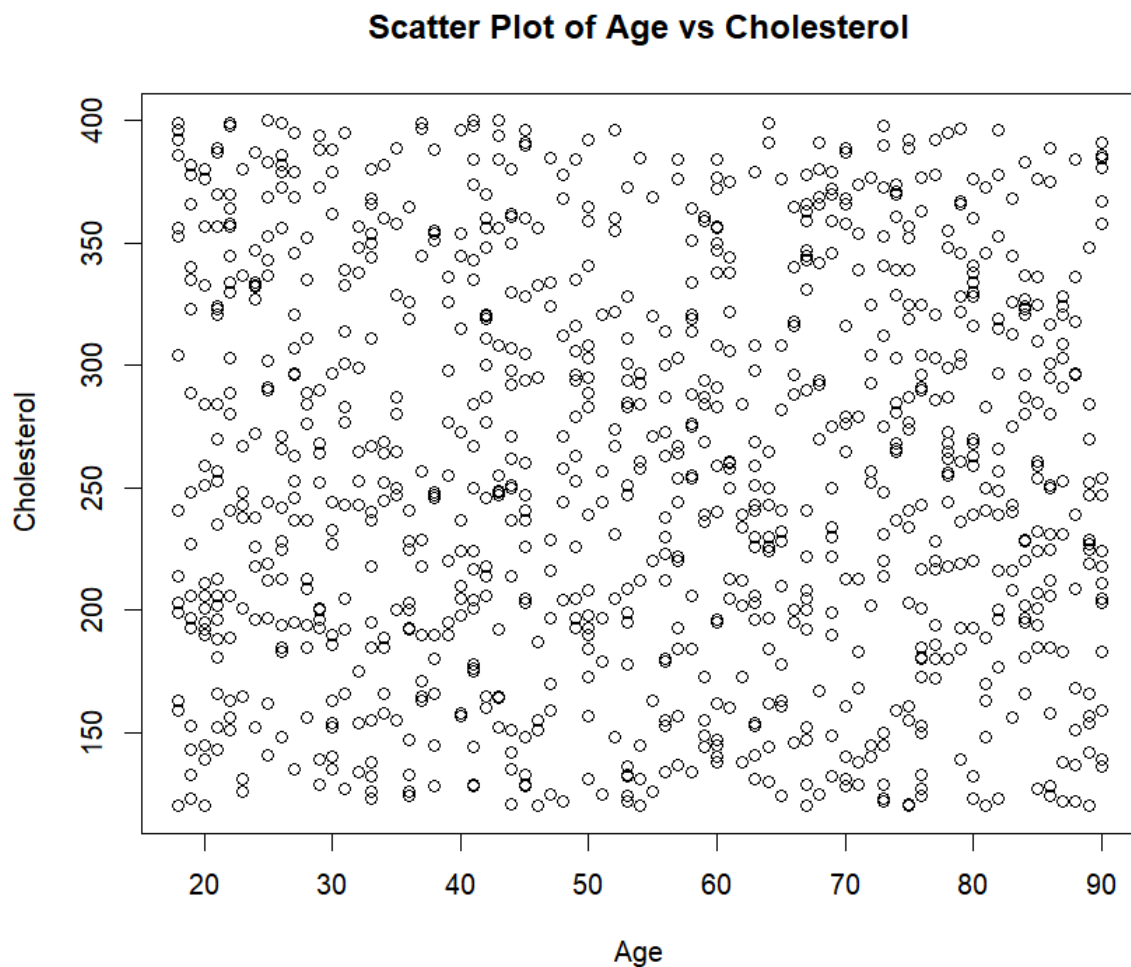


Figure 3.2 Visualise data in a scatter plot using RStudio

1. Calculate the sample correlation coefficient using Pearson's method

The formula for calculating the sample correlation coefficient, r by using the Pearson's technique and the suitable test statistic formula are shown below.

Sample correlation coefficient, r :

$$r = \frac{\sum xy - (\sum x \sum y) / n}{\sqrt{[(\sum x^2) - (\sum x)^2 / n][(\sum y^2) - (\sum y)^2 / n]}}$$

where,

r = Sample correlation coefficient

n = Sample size

x = Value of the independent variable

y = Value of the dependent variable

```
> correlation <- cor(ages, cholesterol, method = "pearson")
> print(paste("Pearson's correlation coefficient: ", correlation))
[1] "Pearson's correlation coefficient: -0.0286713646741576"
```

By using RStudio, we can see that the sample correlation coefficient, r is -0.0287, which indicates that there is a relatively weak negative linear correlation between the age and total cholesterol of patients.

2. Significance Test for Correlation

- **Hypothesis statement:**

Null hypothesis, $H_0 : \rho = 0$ (no linear correlation)

Alternative hypothesis, $H_1 : \rho \neq 0$ (linear correlation exists)

where ρ represents the population correlation coefficient.

- Calculate Test Statistic by Test statistic formula, t:

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

```
> n <- 1000
> r <- correlation
> t_statistic <- r * sqrt((n - 2) / (1 - r^2))
> print(paste("Test Statistic: ", t_statistic))
[1] "Test Statistic: -0.906133556836097"
```

By using RStudio, we get the test statistic, t is -0.906134

Degree of freedom, $df = n - 2$

Find critical value, using $\alpha = 0.05$, $df = n - 2 = 998$

From t-table, since this is a two-tailed test, there are two critical values:

Lower tail critical value $-t_{\alpha/2} = 0.025, df=998 = -1.962344$

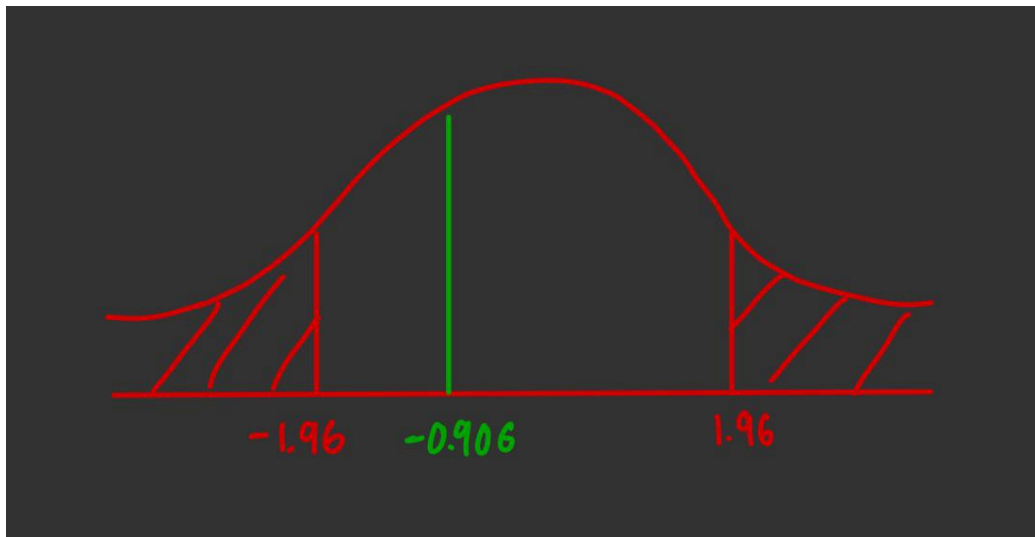
Higher tail critical value $t_{\alpha/2} = 0.025, df=998 = 1.962344$

```
> alpha <- 0.05
> df <- n - 2
> critical_value <- qt(1 - alpha / 2, df)
> print(paste("Critical Value at 95% confidence level: ", critical_value))
[1] "Critical Value at 95% confidence level: 1.96234384621633"
```

From RStudio, the critical value that we obtained is 1.962344

Hence, if the test statistic > 1.962344 or the test statistic < -1.962344 , we will reject the null hypothesis, H_0 . Otherwise, we will fail to reject the null hypothesis, H_0 .

- State the decision:



Since $-1.962344 < t = -0.906134 < 1.962346$ the test statistic falls within the range of the critical values, we **fail to reject** the null hypothesis. In conclusion, there is sufficient evidence to support that there is no linear relationship between the age and total cholesterol of patients, at $\alpha = 0.05$.

Lower tail critical value $-t_{\alpha/2=0.025, df=998} = -1.962344 < \text{test statistic, } t = -0.906134 <$

Upper tail critical value $t_{\alpha/2=0.025, df=998} = 1.962346$

```
> correlation_test <- cor.test(ages, cholesterol, method = "pearson")
> print(correlation_test)
```

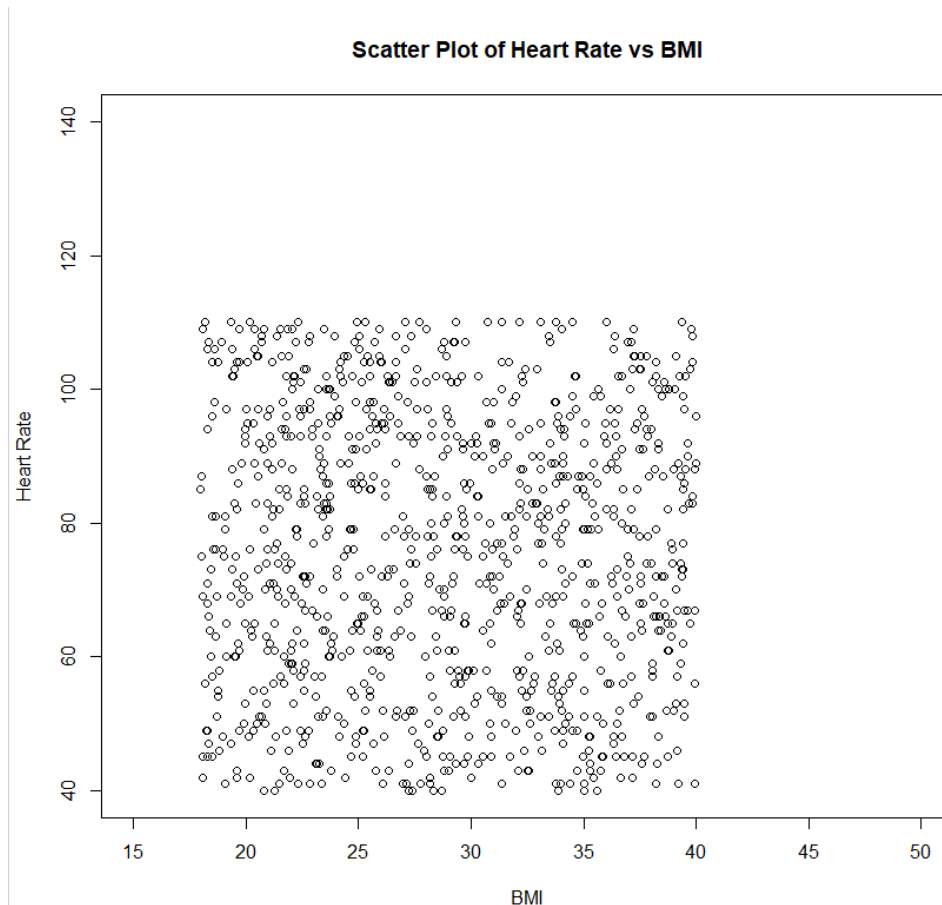
Pearson's product-moment correlation

```
data: ages and cholesterol
t = -0.90613, df = 998, p-value = 0.3651
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.09050358  0.03338105
sample estimates:
      cor
-0.02867136
```

Performing significance test for correlation using RStudio

3.3 Regression Test

In this analysis, the two variables used are BMI and heart rate in which we would test whether the value of heart rate depends on the value of BMI. Thus, BMI is considered as the independent variable (x) while heart rate is considered as the dependent variable (y). In this project, simple linear regression is applied since the regression model is a linear model. The changes in the values of heart rate are based on the changes in the values of BMI.



The Population Linear Regression Model:

$$y = \beta_0 + \beta_1 x + \epsilon$$

Diagram illustrating the components of the Population Linear Regression Model equation $y = \beta_0 + \beta_1 x + \epsilon$:

- Dependent Variable:** y
- Population y intercept:** β_0
- Population Slope Coefficient:** β_1
- Independent Variable:** x
- Random Error term, or residual:** ϵ
- Linear component:** $\beta_0 + \beta_1 x$
- Random Error component:** ϵ

Assumptions:

- Error values (ϵ) are statistically independent and normally distributed for any given value of x

- The probability distribution of the errors is normal and has a constant variance
- The underlying relationship between the x variable and the y variable is linear

1) Estimated Regression Model:

Estimated (or predicted) y value

Estimate of the regression intercept

Estimate of the regression slope

Independent variable

$$\hat{y}_i = b_0 + b_1 x$$

Based on the equation above, b_0 and b_1 are obtained by finding the values of b_0 and b_1 .

The formulas for b_0 and b_1 :

$$b_1 = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

We obtain $b_0 = 78.61554$ and $b_1 = -0.104036$ in R-studio.

```
> cat("b0 (Intercept):", b0, "\n")
b0 (Intercept): 78.61554
> cat("b1 (Slope):", b1, "\n")
b1 (Slope): -0.104036
```

After that, substitute the values of b_0 and b_1 into the estimated regression model:

$$\hat{y}_i = 78.61554 - 0.104036x$$

Based on the above regression model equation, b_0 shows that **78.6155** is the estimated average value of heart rate when the value of BMI is 0 while b_1 shows that **-0.1040** is the estimated change in the average value of heart rate as a result of a one-unit change in BMI.

Perform estimated regression model in R-studio:

```
> model

Call:
lm(formula = Heart.Rate ~ BMI, data = data_subset)

Coefficients:
(Intercept)      BMI
    78.616      -0.104
```

From the image above, we can clearly see the values of the intersection coefficient (y) and slope coefficient (x).

Explained and Unexplained Variation:

$$SST = SSE + SSR$$

Total sum of
Squares

Sum of Squares
Error

Sum of Squares
Regression

$$SST = \sum (y - \bar{y})^2$$

$$SSE = \sum (y - \hat{y})^2$$

$$SSR = \sum (\hat{y} - \bar{y})^2$$

where:

$\bar{y}$ = Average value of the dependent variable

y = Observed values of the dependent variable

$\hat{y}$ = Estimated value of y for the given x value

We obtain SST=403098.3, SSR=438.5837 and SSE=402659.7 in R-studio.

```
> cat("SST (Total sum of squares):", SST, "\n")
SST (Total sum of squares): 403098.3
> cat("SSR (Sum of squares due to Regression):", SSR, "\n")
SSR (Sum of squares due to Regression): 438.5837
> cat("SSE (Sum of squares due to Error):", SSE, "\n")
SSE (Sum of squares due to Error): 402659.7
```

Coefficient of Determinations (R-squared):

$$R^2 = \frac{SSR}{SST} \quad \text{where} \quad 0 \leq R^2 \leq 1$$

We obtain R-squared=0.001088032 in R-studio.

```
> cat("R-squared:", R_squared, "\n")
R-squared: 0.001088032
```

Based on the above result, we can interpret that 0.001088032 of the variation in heart rate is determined by the variation of BMI.

The Standard Deviation of the Regression Slope:

$$s_{b_1} = \frac{s_\varepsilon}{\sqrt{\sum (x - \bar{x})^2}} = \frac{s_\varepsilon}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}}}$$

where:

s_{b_1} = Estimate of the standard error of the least squares slope

$s_\varepsilon = \sqrt{\frac{SSE}{n-2}}$ = Sample standard error of the estimate

We obtain $Sb1=0.09978412$ and $SE=20.08648$ in R-studio.

```
> cat("se (Standard Error of Residuals):", se, "\n")
se (Standard Error of Residuals): 20.08648
> cat("sb1 (Standard Error of Slope):", sb1, "\n")
sb1 (Standard Error of Slope): 0.09978412
```

2) Inference the slope using t-test

■ Hypothesis statement:

$H_0: \beta_1 = 0$ (no linear relationship)

$H_1: \beta_1 \neq 0$ (linear relationship does exist)

- Using $\alpha = 0.05$, $df = n-2 = 998$, we obtain critical values $-t_{0.025, 998} = -1.96$ and $t_{0.025, 998} = 1.96$. We reject H_0 if test statistics < -1.96 or test statistics > 1.96 .

critical value for two-tailed test at alpha = 0.05 and df = 998 : 1.962344

■ Test statistic:

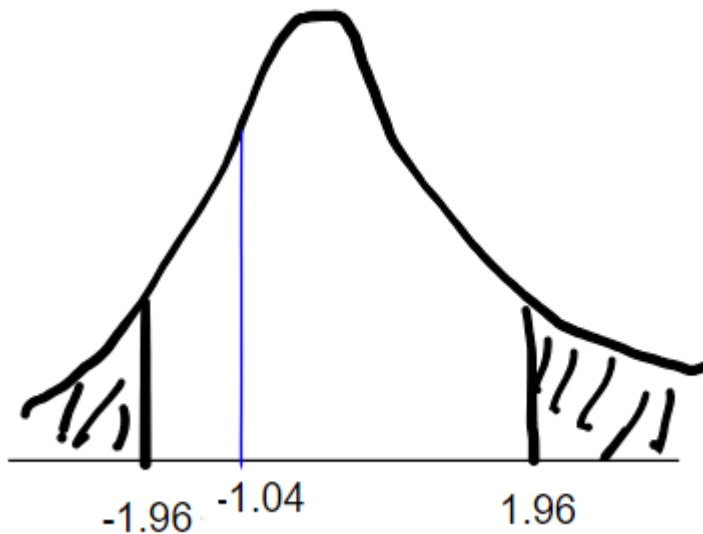
$$t = \frac{b_1 - \beta_1}{s_{b_1}}$$

$$d.f. = n - 2$$

We obtain test statistic, $t_0 = -1.042611$ in R-studio.

```
> cat("t-statistic:", t_statistic, "\n")
t-statistic: -1.042611
```

■ Decision:



Since $-1.96 < t_0 < 1.96$, H_0 is **not rejected**. Thus, at $\alpha = 0.05$, there is insufficient evidence to conclude that a linear relationship exists between heart rate and BMI value.

■ Summary:

```
> summary(model)

Call:
lm(formula = Heart.Rate ~ BMI, data = data_subset)

Residuals:
    Min       1Q   Median       3Q      Max
-36.451 -17.292  -0.419  17.265  35.481

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  78.61554    2.94530   26.692  <2e-16 ***
BMI          -0.10404    0.09978   -1.043    0.297
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.09 on 998 degrees of freedom
Multiple R-squared:  0.001088, Adjusted R-squared:  8.712e-05
F-statistic: 1.087 on 1 and 998 DF, p-value: 0.2974
```

From the summary, we obtain the following informations about this test:

$b_0 = 78.6155$

$b_1 = -0.1040$

$s_{b1} = 0.0998$

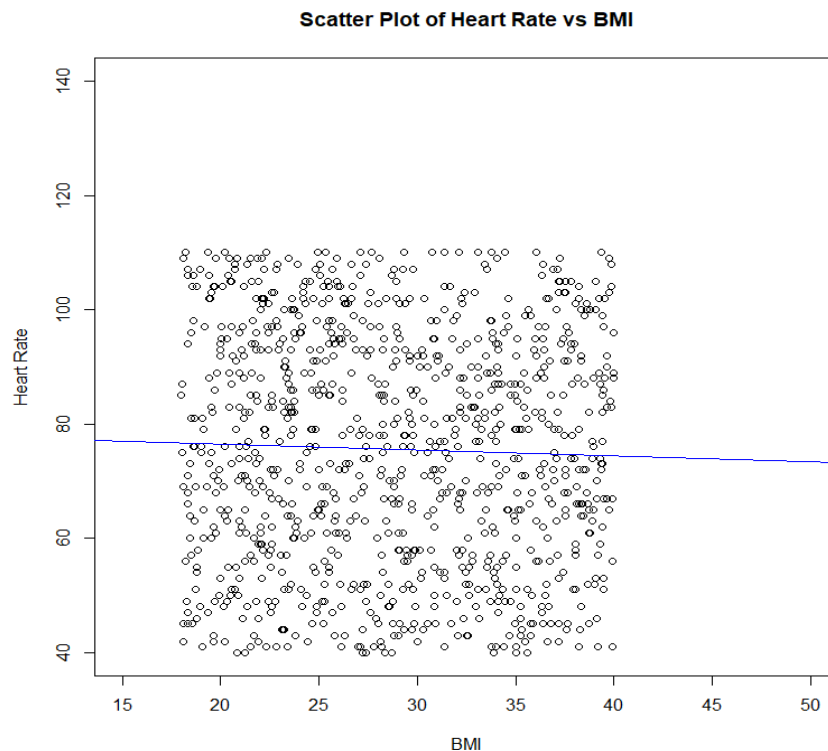
$s_e = 20.09$

$df = 998$

$R\text{-squared} = 0.0011$

$p\text{-value} = 0.2974$

Scatter plot with regression line:



3.4 Chi-Square test of independence

In this analysis, we will use the variables **Sex** and **Obesity** to test the independence of sex and obesity using a Two-Way Contingency Table at a 95% confidence level. As a result, we apply the Chi-Square Test of Independence in conjunction with a two-way contingency table to test for the relation between two qualitative variables.

```
> sex_obesity_table<-table(heart_attack_prediction_dataset$Sex,heart_attack_prediction_dataset$Obesity)
> sex_obesity_table
```

	0	1
Female	184	144
Male	339	333

Observed Frequencies for variables Sex and Obesity

1. Hypothesis

H_0 : Sex and Obesity are Independent

H_1 : Sex and Obesity are Related

2. Find the critical value:

Critical value, $\chi^2 = 3.841$

(with $df = (2-1)(2-1) = 1$, $\alpha = 0.05$)

```
> alpha4<-0.05
> df4 <- (nrow(sex_obesity_table) - 1) * (ncol(sex_obesity_table) - 1)
> critical4=qchisq(alpha4,df=df4,lower.tail = FALSE)
> critical4
[1] 3.841459
```

Critical value, χ^2 using RStudio

By using RStudio, the value of the critical value, χ^2 is 3.841459

3. Calculate the expected counts:

Sex	Obesity				Total
	0		1		
	Obs.	Exp.	Obs.	Exp.	
Male	339	$\frac{672 \times 523}{1000}$ = 351.456	333	$\frac{672 \times 477}{1000}$ = 320.544	672
Female	184	$\frac{328 \times 523}{1000}$ = 171.544	144	$\frac{328 \times 477}{1000}$ = 156.456	328
Total	523	523	477	477	1000

4. Calculate the test statistic value:

Test statistic formula, $\chi^2 = \sum_{all\ cells} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$

where,

o_{ij} = Observed count at row i column j

e_{ij} = Expected count at row i column j

r = total number of rows

c = total number of columns

- Calculate manually based on a formula

Cell, ij	Observed Count, o_{ij}	Expected Count, e_{ij}	$\frac{(o_{ij} - e_{ij})^2}{e_{ij}}$
1,1	339	$\frac{672 \times 523}{1000} = 351.456$	$\frac{(339 - 351.456)^2}{351.456} = 0.441$
1,2	333	$\frac{672 \times 477}{1000} = 320.544$	$\frac{(333 - 320.544)^2}{320.544} = 0.484$
2,1	184	$\frac{328 \times 523}{1000} = 171.544$	$\frac{(184 - 171.544)^2}{171.544} = 0.904$
2,2	144	$\frac{328 \times 477}{1000} = 156.456$	$\frac{(144 - 156.456)^2}{156.456} = 0.992$
χ^2			2.821

- Calculate using RStudio

```
> chi_square_test <- chisq.test(sex_obesity_table)
> print(chi_square_test)
```

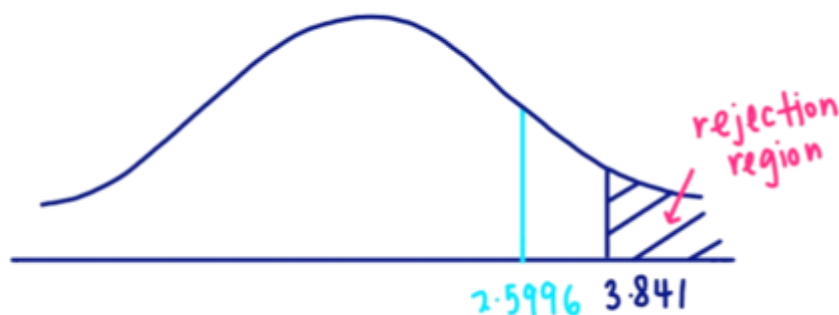
Pearson's Chi-squared test with Yates' continuity correction

```
data: sex_obesity_table
X-squared = 2.5996, df = 1, p-value = 0.1069
```

Test Statistic, χ^2 using RStudio

The value of the test statistic, χ^2 calculated using RStudio is 2.5996 with the df=1 and p-value = 0.1069.

5. Conclude



Since the test statistic value ($\chi^2=2.5996$) < critical value ($\chi^2_{0.05}=3.841$), thus **failed to reject the null hypothesis**, H_0 . There is enough evidence to support the claim that sex and obesity are independent at $\alpha = 0.05$.

4.0 Conclusion

For the 1 sample hypothesis test, we desire to test whether the mean of the patient's age is greater than 50 years old at a 0.05 significance level. We found that the mean of the patient's age is greater than 50 years. Hence, we reject the null hypothesis. There is enough evidence to support the claim that the mean age for the patient is greater than 50 years. From this result, we know that the age of patients is mostly more than 50 years.

Next, for the correlation test, our results indicate that there is insufficient evidence to support a linear relationship between the age and total cholesterol levels of patients at the 0.05 significance level. Hence, we fail to reject the null hypothesis. This means that, based on our sample data, we cannot conclude that age is a significant predictor of total cholesterol levels in patients. Therefore, other factors might need to be considered to explain variations in cholesterol levels among patients.

In the regression test, we came out with the result that BMI values do not affect the heart rate of people. According to the decisions made, we fail to reject the null hypothesis again. Additionally, based on the scatter plot, the regression line is almost straight, reflecting that there is no linear relationship between BMI values and heart rate. Applying the result in real-time situations, this test concludes that people with higher BMI levels do not necessarily have higher heart rates.

Based on the Chi-Square test of Independence, we determined that there is no significant relationship between sex and obesity among 1000 respondents at a significance level of $\alpha=0.05$. Hence, we failed to reject the null hypothesis. This is because the test indicates that both sex and obesity are independent variables, revealing that a person's sex does not influence their likelihood of being obese.

Above all, we can conclude by saying that the opportunity for someone to have a heart attack is higher if the patient's age is greater than 50 years old in terms of the sample hypothesis test. Additionally, based on the results of the correlation test, regression test and chi-squared test, we all fail to reject the null hypothesis meaning that the total cholesterol levels are not fully dependent on age, the heart rates are not fully dependent on BMI levels and obesity is not fully caused by sex.

Carrying out this project is really beneficial to us as we have an opportunity to explore the use of R-studio which is an effective tool to analyse both small and big data. This enables us to get prepared dealing with real-time situations when we have to analyse large amounts and different kinds of data in reality. Lastly, we express our appreciation to our lecturers, Dr Nor Hizan bt Mohamed Radzi for providing meaningful guidance and help throughout this project.

5.0 Appendix

<https://www.kaggle.com/datasets/iamsouravbanerjee/heart-attack-prediction-dataset>

Video presentation link:

<https://youtu.be/ILuqBDdwug8>